The palaeomagnetic age of the WNW-striking dikes around Gothenburg, Sweden

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The remanent magnetization of 10 oriented samples (30 specimens) from a site on the 100—200-m-broad, WNW.-striking dolerite dike at Tuve near Gothenburg in south-western Sweden has been measured after stepwise demagnetization in an alternating magnetic field of up to 1400 Oe. A stable, supposedly primary component of the remanent magnetization was isolated at $F_{max} = 500$ Oe, revealing a north-palaeopole position of 17°N., 59°E. (δ_p, $\delta_m = 2°$, 3°).

A comparison with other known pole positions of "stable" Europe makes the dike about 800–900 m. y. old, corresponding to a Sveconorwegian (Dalslandian) age.

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The purpose of the present paper is to make a palaeomagnetic contribution to the age problem of the WNW.-striking dikes of the Gothenburg region, the age of which is not yet known for certain.

In the coastal regions around Gothenburg in south-western Sweden, two generations of basic dikes are found (Lundegårdh 1953, 1957), the members of the older generation striking NNE. to NE., while those of the younger group strike WNW. (Ljungner 1927, Lundegårdh 1953, 1957). Both types are only known in the Gothic part of the Precambrian basement, and no direct datings.

Johansson (1931) correlated the older generation of dikes, which may be up to a few metres in width, with the diabase dikes on the Koster islands, situated about 130 km N. of Gothenburg. The Koster dikes are older than the Bohus granite (Gavelin 1914), and it seems that these dikes were formed at the same time as the greenstones of the Dal formation (Lundegårdh 1957).

The younger WNW-striking dikes attain a width of up to about 200 m (the Tuve dike), and the age is not known accurately. However, although the strong, late-deuteric alterations of the feldspars, which are considered typical of the NW.-striking dolerites of northern Dalsland, are not typical of the younger dikes around Gothenburg, Larsson (1956) considers a Dalslandic age probable for them also, suggesting a genetic relationship between the metabasites of the Dal formation and the Gothenburg dikes.

For further discussions of the Gothian–Pregothian correlation problems and absolute ages, see Geijer (1963), Gorbatschev (1971) and Welin (1966).

In contrast to this, Lundegårdh (1953, 1957) suggests either an early Palaeozoic or possibly a Jotnic age, because of the resemblance with the Jotnic Åsby diabase. A Permian age is unlikely, because the diabases and rhomb porphyries, which stretch far southward from the Oslo region into Bohuslän and are known to occur as closely as about 50 km N. of Gothenburg (Asklund 1947, Samuelsson 1971), all strike north–south. Furthermore, palaeomagnetic investigations of these dikes give quite different pole positions from those in the present investigation (Abrahamsen & Thorning, in prep.).

Palaeomagnetic measurements on the Tuve dike

From a single locality on the big Tuve dike, 10 independently orientated hand samples were collected by dr. Lennart Samuelsson and kindly placed at the author's disposal for further investigation.

The site is located 2.8 km NNW. of Björlanda Church on the coast, where the biggest of the younger dikes, the Tuve dike, which at this locality is between 80 and 120 m broad, crops out. The site falls within 40 and 60 m from the southern contact. The northern contact is not exposed, but the site is obviously close to the centre of the dike.

Ten fresh and unweathered samples were loosened with a hammer and put back again into place, and levelled surfaces were moulded on them in plaster of Paris, on which the magnetic north was drawn. No local magnetic deviations could be demonstrated (accuracy within ± 2°), while the regional magnetic variation was 2° W. In the laboratory, finally, three cylindrical specimens (2.5 cm × 2.5 cm) were drilled from each of the 10 samples.

The direction and intensity of the natural remanent magnetization (NRM) were measured on a Molyneaux low-speed spin magnetometer, the range of the intensity of the NRM being between 1.5 and 3.7 × 10^{-3} emu/cm³.

To save time, first one pilot specimen from each sample was a.f. demagnetized within a field-free space from Helmholtz coils (± 10γ) and afterwards re-measured; the maximum value of the stepwise increased field being F_{max} = 1400 Oe. In Table 1, the results from these 10 pilot specimens are given, together with the conventional Fisher statistics on a sphere (for further explanation, consult Irving 1964).

As will be seen in Fig. 1, there is a pronounced minimum of 2.1° in the value of a_{95} (the radius of the 95 % circle of confidence) at a demagnetizing maximum field of F_{max} = 500 Oe. At this value of the demagnetizing field, 30-40 % of the original NRM was still left (Fig. 2), and even at F_{max} = 1400 Oe, the ratio between the demagnetized and the original remanent magnetization J_{1400}/J_0 was between 8 and 11 %, indicating that the NRM includes a significant amount of stable components of high coercivities.

On the basis of this information from the pilot specimens, all the remaining specimens were demagnetized and re-measured with F_{max} = 400, 500 and 600 Oe respectively. The results are given in Table 2 as mean directions of each sample

Table 1. Pilot specimens.

F_{max} (Oe)	N	D	I	a_{95}	k	R
0	10	113.8°	68.3°	17.9°	8.1	8.90164
25	10	111.4	61.0	4.5	115.0	9.92177
50	10	114.0	56.3	5.1	90.3	9.90043
100	10	117.5	52.7	4.3	124.8	9.92788
150	10	118.8	53.7	3.6	178.6	9.94963
200	10	118.6	55.2	3.1	239.4	9.96241
250	10	118.3	56.0	2.7	300.9	9.97009
300	10	118.6	56.6	2.3	436.5	9.97938
400	10	119.3	56.4	2.2	436.5	9.98058
500	10	118.7	56.3	2.1	504.5	9.92216
600	10	118.3	56.1	2.6	344.2	9.97385
800	10	116.7	56.0	4.4	119.9	9.92495
1000	10	116.0	55.9	6.1	63.1	9.85759
1400	10	110.0	55.0	11.6	18.0	9.50178

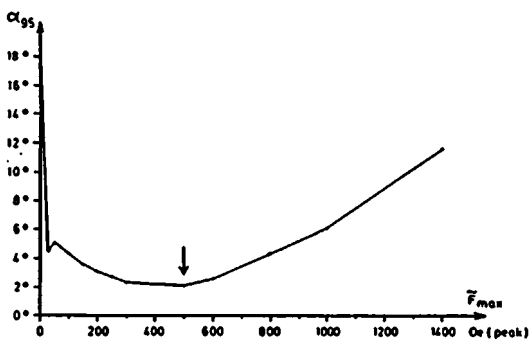


Fig. 1. Minimum in the 95 % circle of confidence for 10 pilot specimens, as found at a demagnetizing maximum field strength of F_{max} = 500 Oe.

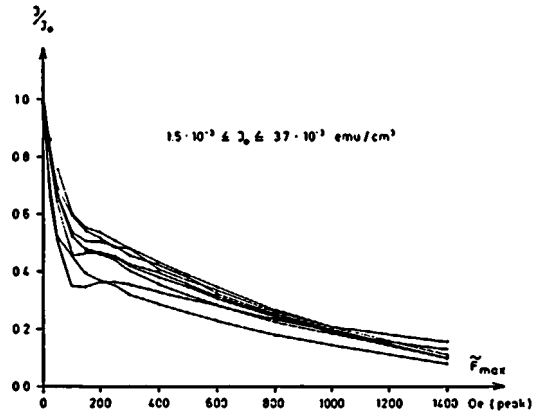


Fig. 2. Intensity curves (normalized) for eight pilot specimens.

Table 2. Mean directions of samples (unit weight to specimens).

Sample no.	F_{max} (Oe)	N	D	I	a_{95}	k	R
1	NRM	3	112.3°	65.1°	5.6°	473.1	2.99577
2	NRM	3	117.3	66.6	2.8	1934.9	2.99896
3	NRM	3	113.4	71.5	2.6	2126.5	2.99905
4	NRM	3	124.3	69.3	3.8	1422.5	2.99859
5	NRM	3	85.9	67.0	5.8	439.4	2.99544
6	NRM	3	114.4	66.8	5.1	563.2	2.99644
7	NRM	3	112.4	69.1	3.3	1317.8	2.99848
8	NRM	3	129.2	61.5	14.4	73.4	2.97275
9	NRM	3	98.1	67.9	6.8	327.2	2.99388
10	NRM	3	98.5	62.1	6.6	341.7	2.99414
1	400	3	121.0	54.2	2.8	1816.8	2.99889
2	400	3	121.4	58.0	4.4	763.7	2.99738
3	400	3	116.6	50.9	5.9	437.0	2.99542
4	400	3	120.2	60.4	2.6	2086.7	2.99904
5	400	3	117.3	53.5	4.4	758.4	2.99736
6	400	3	122.9	56.2	4.7	672.7	2.99702
7	400	3	125.8	58.0	6.5	353.4	2.99434
8	400	3	124.7	53.8	2.8	1818.0	2.99889
9	400	3	113.0	75.0	6.3	378.3	2.99471
10	400	3	111.9	54.4	7.0	308.6	2.99351
1	500	3	121.2	54.9	3.7	1108.9	2.99819
2	500	3	121.9	57.9	4.4	773.7	2.99741
3	500	3	115.6	52.4	3.2	1430.8	2.99860
4	500	3	115.4	57.9	5.2	547.1	2.99634
5	500	3	119.5	55.7	0.7	28850.8	2.99993
6	500	3	124.5	57.0	4.0	932.2	2.99785
7	500	3	125.0	58.0	5.4	511.1	2.99608
8	500	3	125.4	54.0	3.2	1407.6	2.99857
9	500	3	113.0	57.2	5.5	496.6	2.99597
10	500	3	112.4	54.9	4.7	687.9	2.99709
1	600	3	119.3	54.7	2.9	1706.7	2.99882
2	600	3	124.9	58.1	3.5	1231.0	2.99837
3	600	3	113.2	50.3	2.1	3445.5	2.99941
4	600	3	110.2	55.9	8.4	212.5	2.99058
5	600	3	121.1	56.5	2.5	2401.1	2.99916
6	600	3	125.3	57.6	4.6	714.9	2.99720
7	600	3	128.4	57.5	6.4	363.2	2.99449
8	600	3	124.4	53.3	2.6	2150.2	2.99906
9	600	3	111.8	55.9	7.6	260.3	2.99231
10	600	3	114.3	55.1	5.3	541.9	2.99630

(three specimens) with unit weight to specimens.

Finally, in Table 3 the statistics on site level are given with unit weight to samples, based on the values of Table 2. The best value for the direction of the stable component of the supposed primary magnetization is again found for F_{max} = 500 Oe, with a_{95} = 2.0°, N = 10, precision parameter k = 575, mean declination D_m = 119° and mean inclination I_m = + 56°.

With the present geographical co-ordinates for the site (57.786°N., 11.811°E.), this corresponds in the present grid to a palaeopole position of 17°N., 59°E. for the north pole (or 17°S., 121°W. for the south pole).

The stable direction found is not far from the NRM direction, as shown in Fig. 3, and both directions are significantly different from the present direction of the geomagnetic field at the site. The NRM direction is intermediate between the present field and the stable component found; thus, most of the soft viscous components of the NRM were probably acquired in quite recent time, induced by the present geomagnetic field.

Thermomagnetic curves (susceptibility versus temperature) of rock powder (diameter less than 250 μ) from samples Nos. 3, 7 and 10 are shown in Fig. 4. The Curie points are found between

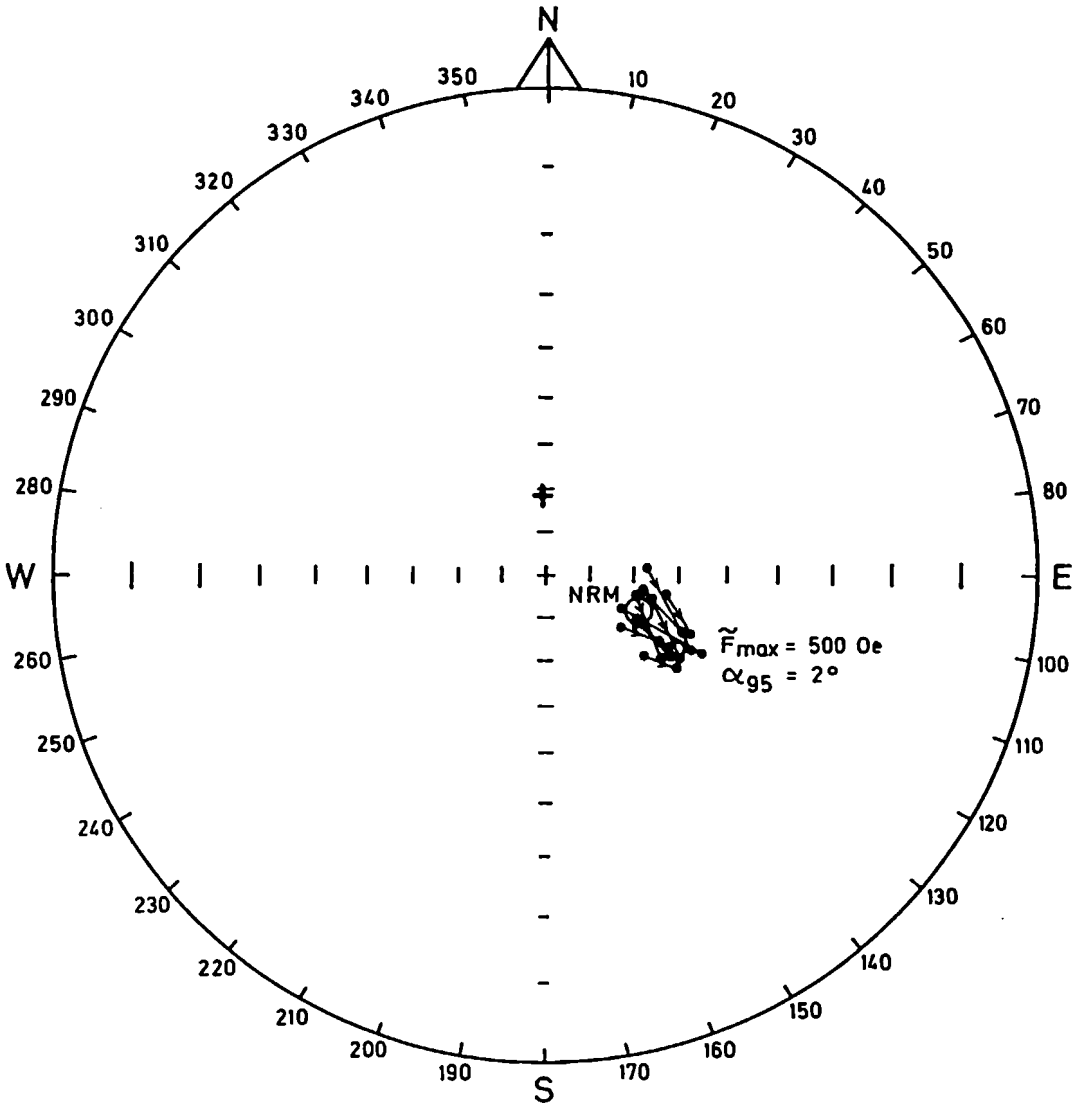


Fig. 3. Mean directions of the 10 samples (each a mean of three specimens) before and after a. f. demagnetization, with the corresponding 95 % circles of confidence. The cross indicates the direction of the present geomagnetic field.

Table 3. Mean direction of site, with pole position (unit weight to samples).

F_{max} (Oe)	N	D_m	I_m	a_{95}	k	R	*Pole position*	δ_p	δ_m
NRM	10	110.6°	67.1°	3.6°	178.7	9.94966	31.6°N., 57.0°E.	5.0°	5.9°
400	10	119.4	55.7	2.2	444.6	9.97976	16.7°N., 59.0°E.	2.4	3.2
500	10	119.3	56.0	*2.0*	574.8	9.98434	*17.0°N., 58.9°E.*	2.2	2.9
600	10	119.1	55.6	2.5	346.6	9.97403	16.7°N., 59.3°E.	2.7	3.6

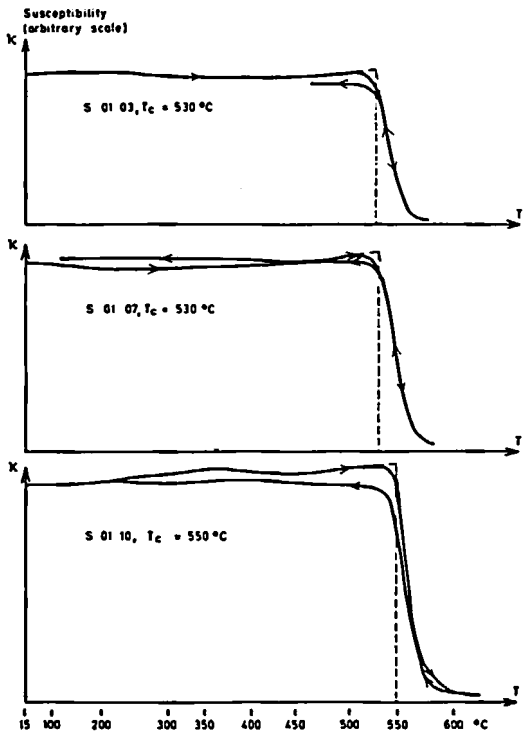


Fig. 4. Thermomagnetic curves of three powder specimens, indicating Curie points between 530°C and 550°C, typical of nearly pure magnetite.

530° and 550°C, which corresponds to an (assumed) stoichiometric composition of titanomagnetite $Fe_{3-x}Ti_xO_4$ with $x \simeq 4$–7 mol-% or a fairly pure magnetite.

Thin sections show that the essential minerals are zonar plagioclase, which may be somewhat sericitized with centres stained by reddish-brown pigments, brown augite and homogeneous subhedral magnetite (about 1–3 vol. %) with a grain size usually between 0.2 and 2 mm; accessories are olivine, which is partly altered and with rims and small grains of magnetite, and a little biotite.

The carriers of the softer parts of the remanent magnetization with coercivities about 100 Oe (cf. Fig. 2) are without doubt the subhedral magnetites, but, as they are all poly-domaine, they cannot account for the high coercivities up to and above 1400 Oe also present. However, submicroscopic single-domaine magnetite grains may be supposed to occur within the altered olivines, and in the case of elongated shapes, the coercivities may be quite high (Strangway *et al.* 1968).

The small ore minerals in the olivines are supposed to have been formed by deuteric alterations shortly after the solidification of the magma. The stable part of the remanent magnetization should therefore strictly be considered a chemical remanent magnetization (CRM) rather than a TRM. But as the creation of this secondary remanent component was probably close in time to the primary TRM, the directions of the two should be expected to be close to each other.

Cooling down below the blocking temperature, which forms an interval rather than a point because of the different shapes, sizes and compositions of the ferromagnetic minerals, is a relatively slow process in a dike of these dimensions. With a width of $D = 100$ m, a temperature interval of crystallisation for the magma as a whole of T_2–$T_1 = 1000°$–$800°C$ and a specific heat of solidification of $L = 100$ cal/g, it may be estimated (Fig. 2 in Jaeger 1957), that it will take about 100 y for the central part of the dike to cool down from 600°C to 500°C, and somewhat longer to cool from 500°C to 400°C. The presence of a palaeosecular variation at that time, with an amplitude of 10–15 degrees, and a period of a few hundred years (as for the present geomagnetic field) would to a certain extent smooth out the blocked directions, so that the present stable direction would not be a true spot-reading in time, even in the absence of a later-formed CRM.

To be considered a palaeomagnetic pole in the usual sense (Creer 1970), the pole should originate as a smoothed mean from rocks formed over a rather long period of about 10^5–10^6 y, thus smoothing out secular variations of non-dipole shape. As the present data originate from only one site, this condition is evidently not fulfilled; on the other hand, it may also be questioned whether the pole should be considered a virtual geomagnetic pole in the sense of a true spot-reading in time, because of the possible effects of cooling time and deuteric alterations.

In the present context, however, where dating is the main purpose, it appears reasonable to suppose that the pole position found may be compared with other truly smoothed palaeomagnetic poles, to give a palaeomagnetic estimate of the age of the dike and hence of the WNW.-striking dike swarm as a whole.

Comparison with other palaeomagnetic poles

In Table 4, selected data for Precambrian palaeomagnetic poles for "stable" Europe younger than about 1400 m. y. with their 95 % polar ovals of confidence are compiled, essentially according

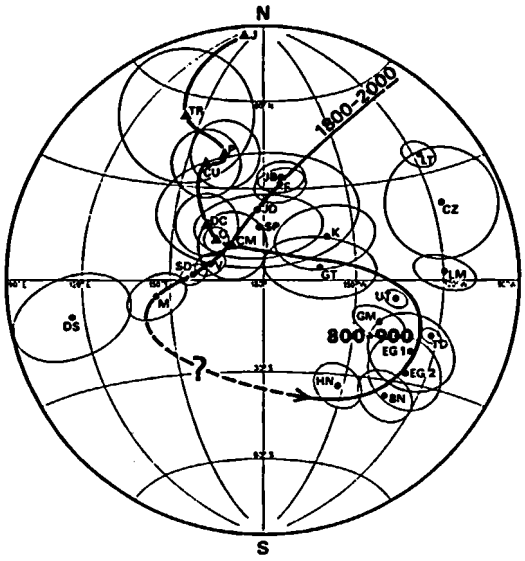


Fig. 5. Palaeomagnetic poles and suggested polar wandering curve (after Spall (1973) and Creer (1970)). The position of the pole of the Tuve dike (TD) suggests an age of about 800–900 m.y.

to Spall (1973). Further, some post-Precambrian mean poles with their circular standard deviations are quoted from Creer (1970). These data are plotted in Fig. 5, together with the polar wandering curve for "stable" Europe, as partly suggested by Neuvonen (1970) and further elaborated by Spall. Some of the Precambrian poles do not fit the polar curve well, as is discussed in some detail by Spall. The Lower Torridonian (LT) and Czechoslovakian (CZ) poles may eventually require one more loop on the pole curve between 1400 and 900 m.y., but the data do not yet appear sufficiently to support this more complicated solution. The K/Ar ages of the five Swedish hyperite dolerites (HD) between 781 and 1573 m.y. may not reflect their palaeomagnetic age, which is suggested to be 800–900 m.y. (Spall). Mulder (1971, Priem *et al.* 1968) suggested that the palaeomagnetic directions of the Swedish Jotnian basalts (JB) and late Jotnian dolerites (JD) were caused by a Caledonian remagnetization, as their poles are close to the known Palaeozoic poles of Europe. This hypothesis of remagnetization, however, does not appear to be necessary, as Spall pointed out, because the poles of JB and JD fit well into the 1400-m.y. segment of the pole path. Finally, the trend of the late Precambrian segment is in good agreement with the two upper Precambrian poles for eastern Greenland (GM and GT).

Now, comparing the position of the pole of the Tuve dike (TD) with the other poles in Fig. 5, it must be concluded that the magnetic age of the Tuve dike is neither close to 1400 m.y. (i. e. Jotnian) nor Palaeozoic but must be about 800–900 m.y., that is of Sveconorwegian (Dalslandian) age comparable in palaeomagnetic age with the Egersund dolerites (EG) in southern Norway and presumably the hyperite dolerites (HD) in southern Sweden. Thus the hypothesis of Larsson (1956), as discussed in the introduction, is supported by the present investigation.

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Table 4. Middle, late and post-Precambrian poles for "stable" western Europe (and Greenland).[a]

Legend (Fig. 4)	Rock unit	Country	Pole Long.	Pole Lat.	δ_p	δ_m	Age	Reference
UT	Upper Torridonian sst	Scotland	137°W.	6°S.	4°	6°	805 m.y., Rb/Sr isochron[c]	Irving *et al.* 1957, Moorbath 1969
LT	Lower Torridonian sst	Scotland	118 W.	35 N.	5	8	990 m.y., Rb/Sr isochron[c]	Irving *et al.* 1957, Moorbath 1969
LM	Longmyndian sediments	England	120 W.	2 N.	7	13	600—700 m.y., K/Ar	Creer 1957; Wright 1969
CZ	Sediments	Czechoslovakia	117 W.	21 N.	21	21	Eocambrian	Bucha 1965
GT	Tillite series	Greenland	161 W.	4 N.	10	19	Eocambrian	Bidgood *et al.* 1961
GM	Multi-coloured series	Greenland	161 W.	13 S.	5	10	Upper Precambrian	Bidgood *et al.* 1961
SP	Sparagmites	Norway	179 E.	18 N.	12	25	Eocambrian	Harland *et al.* 1959
BN	Basement rocks	Norway	133 W.	36 S.	10	12	850—950 m.y., Rb/Sr. K/Ar, U/Pb	Poorter 1972
HN	Hunnedalen dikes	Norway	152 W.	34 S.	9	10	<850—950 m.y.	Poorter 1972
EG1	Egersund dikes	Norway	128 W.	28 S.	15	18	<850 m.y.	Storetvedt *et al.* 1968
EG2	Egersund dikes	Norway	129 W.	22 S.	14	16	850—950 m.y.	Poorter 1972
JB	Jotnian basalts	Sweden	174 W.	32 N.	5	10	745—931 m. y., K/Ar	Priem *et al.* 1968
JD	Late Jotnian dolerites	Sweden	178 E.	23 N.	20	40	Late Jotnian	Priem *et al.* 1968
HD	Hyperite dolerites	Sweden	134 W.	12 S.	17	23	781—1573 m.y., K/Ar	Priem *et al.* 1968
DS	Dolorites and basalts	Sweden	115 E.	10 S.	16	28	1200 m.y., K/Ar, Jotnian	Dyrelius 1970
SD	Satakunta dolerites	Finland	158 E.	2 N.	3	5	970—1330 m.y., K/Ar	Neuvonen 1965
V	Vaasa dolerites	Finland	164 E.	7 N.	3	6	? Jotnian	Neuvonen 1966
M	Märket dolerite	Finland	146 E.	6 S.	7	11	Younger than Rapakivi	Neuvonen *et al.* 1969
F	Föglö dolerite	Finland	174 W.	31 N.	6	12	? Older than Rapakivi	Neuvonen *et al.* 1969
K	Kumlinge dolerites	Finland	159 W.	13 N.	11	19	?	Neuvonen 1970
Cm		Western Europe	170 E.	11 N.	11[b]		Cambrian	Creer 1970
O		Western Europe	165 E.	13 N.	4		Ordovician	Creer 1970
D-C		Western Europe	161 E.	18 N.	10		Pevonian to Middle Carboniferous	Creer 1970
Cu		Western Europe	157 E.	39 N.	11		Upper Carboniferous	Creer 1970
P		Western Europe	163 E.	41 N.	12		Permian	Creer 1970
Tr		Western Europe	142 E.	53 N.	22		Triassic	Creer 1970
J		Western Europe	131 E.	83 N.	—		Jurassic	Creer 1970

a Precambrian data essentially selected from Spall's Table 1 (1973).
b Circular standard deviation of distribution of formation-mean poles.
c The original Rb/Sr ages are re-calculated using $\lambda_{Rb} = 1.39 \cdot 10^{-11}$ y^{-1} (instead of $1.47 \cdot 10^{-11}$), thus increasing the ages by 6 %.

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